

Reliability Breakdown of Wind Turbine Operating Reserve Rules under Stale SCADA Telemetry and Boundary-Risk Posterior Diagnostics

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Abstract—Operating reserve screening near the demarcation boundary between maximum power point tracking (MPPT, Region 2) and blade-pitch regulation (Region 3) is critical for imbalance risk mitigation and frequency security in wind-integrated power systems. However, industrial Supervisory Control and Data Acquisition (SCADA) telemetry is routinely afflicted by communication queuing delays, packet serialization dropouts, and unobservable pitch registers. While deterministic aerodynamic power-curve quantile rules achieve the lowest PSREI reserve-screening cost under pristine telemetry (589,535 kW · h at immediate dispatch $h = 1$ and 881,367 kW · h at 1-hour dispatch $h = 6$, with $\sim 6.8\%$ violation), they exhibit systematic reliability breakdown under communication latency. Under a 6-step (60 min) synthetic transmission lag, the steep change in local control sensitivity near rated wind speed amplifies state-state estimation errors, driving physical rule violation rates to $24.0\% \pm 1.7\%$ ($h = 1$) and $12.2\% \pm 1.2\%$ ($h = 6$), severely breaching the nominal 10% violation target ($q^* = 0.90$), corresponding to 127.6 MWh of unhedged shortfall exposure in the balancing-risk proxy. To defend against telemetry staleness, we formulate a Cyber-Physical Operational Boundary Framework under Information Freshness Constraints, coupling time-varying directed wake graphs with multi-task boundary-risk posteriors that exploit non-pitch electromechanical transients for blind-spot state reconstruction. Evaluated across a five-seed benchmark on the 134-turbine WTB plant under strictly frozen validation calibration, state-conditional recalibration recovers physical violation to 10.7% (cutting shortage to 57.0 MWh), while joint spatio-temporal posteriors further compress shortage to 53.9 MWh (a combined 57.7% reduction) and double boundary recall over collapsed physical rules (0.417 vs. 0.196). Capacity-matched dense and routed architectures achieve statistical parity under clean telemetry (cost ratio 1.045, $p = 0.380$), while decoupled modular representations (Frozen Backbone + Residual Quantile) achieve lower reserve cost (1.284M kW · h) and compliant tail coverage (9.7% violation). Withheld-channel counterfactual stress testing confirms robust regime recovery under unobservable pitch (NMI 0.561), and Point of Common Coupling (PCC) aggregation demonstrates net plant-level portfolio smoothing. These empirical boundaries establish an accountable pre-dispatch risk diagnostic framework for cyber-resilient grid reserve management, delineating operational trade-offs between unadapted physical rules, state-conditional recalibration, and learned spatio-temporal posteriors under telemetry degradation.

Index Terms—Wind turbine operating reserve; SCADA telemetry degradation; transmission latency; aerodynamic power curve; reliability breakdown; boundary-risk posterior; selective decision abstention; MPPT-to-pitch transition.

I. INTRODUCTION

In bulk power system balancing operations, real-time generation deficits incur severe asymmetric shortage penalties relative to surplus energy; consequently, quadratic loss metrics like root-mean-square error (RMSE) mask catastrophic tail shortfall events that govern system frequency security and operating reserve adequacy [1], [2], [3], [4], [5]. Wind-farm operating reserve screening near the demarcation boundary between maximum power point tracking

(MPPT, Region 2) and blade-pitch regulation (Region 3) represents an acute operational bottleneck. At this demarcation boundary, the governing aerodynamic control law undergoes a structural discontinuity: local power sensitivity drops abruptly from a cubic trajectory ($\partial P / \partial v \propto v^2$) to zero as active pitch regulation sheds excess aerodynamic lift [6], [7]. Deterministic power-curve formulations embody an intrinsic vulnerability near rated inflow velocities: minor unobserved aerodynamic perturbations cause piecewise control laws to trigger delayed or premature pitch actuation, precipitating severe unhedged reserve shortfalls.

In commercial wind plant operations, Supervisory Control and Data Acquisition (SCADA) telemetry streams across distributed turbine networks face unavoidable packet serialization overhead, gateway queuing delays, and asynchronous timestamp skews [8], [9]. In the commercial Kelmarsh plant, for instance, 99.6% of turbine status transition events carry timestamps falling strictly between 10-minute SCADA grid boundaries, demonstrating that operational state transitions regularly unfold within reporting blind spots. To systematically stress-test reserve vulnerabilities under severe communication backlog, buffer congestion, and gateway dropouts [9], [10], [11], this study evaluates controlled synthetic transmission latency profiles from 10 to 60 minutes. Furthermore, in third-party aggregator, Virtual Power Plant (VPP), and Transmission System Operator (TSO) pre-dispatch triage, primary blade-pitch registers are frequently unobservable due to commercial boundaries and proprietary OEM protocol firewalls. Feeding stale or missing sensor measurements into deterministic physical mappings precipitates severe misclassification, compelling pre-dispatch algorithms to clear reserve margins against obsolete operating points.

Existing operational paradigms treat deterministic physical curves and data-driven learning models as mutually exclusive alternatives, overlooking their complementary operational boundaries under communication stress. Under pristine SCADA telemetry, continuous physical quantiles calibrated to turbine aerodynamic specifications achieve the lowest reserve-screening cost among evaluated methods (589,535 kW · h at immediate dispatch $h = 1$ and 881,367 kW · h at 1-hour dispatch $h = 6$, with $\sim 6.8\%$ violation). However, deterministic physical rules exhibit systematic reliability breakdown under telemetry staleness: delayed anemometer and pitch signals cause severe over-extrapolations near rated wind speed. Under a 6-step (60 min) latency stress test, clean-calibrated physical rule violation rates surge to $24.0\% \pm 1.7\%$ at $h = 1$ and $12.20\% \pm 1.25\%$ at $h = 6$, severely breaching the nominal 10% violation target ($q^* = 0.90$) and leaving 127.6 MWh of unhedged shortfall exposure under the balancing-risk proxy. Conversely, shallow tabular models (Missingness-Aware GBDT) capitalize on ultra-short lag autocorrelation at $h = 1$ but undergo severe breakdown across dispatch-grade horizons ($h = 6$), inflating reserve costs by 13.44% to 29.15% ($p < 0.0001$) because tabular baselines cannot capture spatial wake advection propagating across turbine arrays [12].

To resolve this operational tension, we establish a **Cyber-Physical Operational Boundary Framework under Information Fresh-**

ness Constraints. Rather than asserting universal machine-learning superiority, the framework formalizes the operational division of labor: deterministic physical rules deliver cost-optimal reserve screening under fresh telemetry, whereas learned spatio-temporal representations provide essential cyber-resilience under degraded communication. By extracting secondary electromechanical signatures (active power fluctuations, reactive dynamics, and dynamic wake geometry), learned posteriors preserve operational regime awareness and curtail severe shortage volumes, even when primary pitch channels are unobservable. Evaluated across five random seeds on the 134-turbine WTB benchmark under strictly frozen validation calibration, capacity-matched Dense and Mixture-of-Experts (MoE) heads achieve statistical parity in degraded reserve screening, while decoupled modular models achieve lower total cost and compliant tail coverage, establishing that diagnostic reserve gains stem from shared spatio-temporal representations and decoupled residual quantile calibration rather than complex routing mechanisms.

This paper delivers four bounded, verifiable contributions: 1. **Quantification of Physical Rule Breakdown under Telemetry Staleness:** We expose the systematic reliability collapse of deterministic aerodynamic quantile rules under SCADA communication latency. Across 5 random seeds on the 134-turbine WTB plant, a 6-step (60 min) latency drives clean-calibrated physical violation rates from 6.8% to $24.0\% \pm 1.7\%$ at $h = 1$ and 12.2% at $h = 6$, severely breaching the nominal 10% violation target. 2. **Decisive Multi-Horizon Benchmarking and MoE De-mystification:** Under causal telemetry feeds with validation calibration frozen prior to testing, physical rules achieve the lowest PSREI reserve-screening cost under clean telemetry ($589,535 \text{ kW} \cdot \text{h}$ at $h = 1$, $881,367 \text{ kW} \cdot \text{h}$ at $h = 6$). Under stale telemetry, state-conditional calibration absorbs the primary reliability loss (10.7% violation, 57.0 MWh shortage), while learned posteriors achieve $10.6\% \pm 1.1\%$ violation and 53.9 MWh shortage. Decoupled modular architectures (Frozen Backbone + Residual Quantile) achieve lower reserve cost ($1.284 \text{M kW} \cdot \text{h}$) and compliant tail coverage (9.68% violation), demonstrating that operational gains arise from spatio-temporal feature representations and decoupled residual calibration rather than dynamic MoE routing. 3. **Disentangled Diagnostics and Safeguard Boundary Delineation:** Disentangling calibration adaptation from neural representation reveals that state-conditional calibration recovers 95.8% of recoverable shortage energy ($127.6 \rightarrow 57.0 \text{ MWh}$, a 55.3% reduction), while learned posteriors contribute an additional 5.4% reduction ($57.0 \rightarrow 53.9 \text{ MWh}$, combined 57.7%) and double regime recall over collapsed physical rules (0.416 vs. 0.196 under Delay-6). While condition-matching recalibration restores fleet-average violation below the 10.0% target (with physics reaching 1.229M and 9.46%), inter-seed compliance is strictly bounded at 3/5 seeds (60%). We evaluate selective decision abstention against random and uniform margin controls, demonstrating that heuristic entropy scores fail to isolate tail risk and that uniform margin inflation Pareto-dominates selective rejection under missing pitch. 4. **Delineation of Empirical Multi-Farm Deployment Boundaries:** We establish empirical deployment envelopes across commercial wind plants: architectural adaptability holds under local chronological retraining on ENGIE La Haute Borne (routing NMI = 0.941, ARI = 0.971), but direct zero-shot cross-farm transfer exhibits directional asymmetry (Kelmars → Penmanshiel NMI 0.752–0.770 vs. Penmanshiel → Kelmars 0.341–0.505, pooled 0.557). All evaluated reserve costs represent an upstream Level 1 PSREI operational risk proxy ($\rho = 10$). # Related Work

A. From average wind-power accuracy to transition-window risk

Wind-power forecasting has progressed from site-specific statistical models to spatio-temporal architectures that represent ramps, uncertainty and spatial coupling [1], [3], [5]. Graph models provide a strong accuracy reference by propagating information across the network [13], [14], [15]. Wind-specific studies have further introduced wake-aware graphs, SCADA integration and physics-guided constraints [12], [16], [17], [18], [19]. These advances improve prediction, but they do not by themselves expose the active turbine-

control law to a reserve planner.

MoE routing can represent local response laws, but prediction loss alone does not determine whether the resulting partition has operating meaning [20], [21]. Physics-guided machine learning supplies general principles for introducing scientific constraints, while wind-turbine operating states remain limited by SCADA observability [8], [22], [23]. The MPPT-to-pitch transition itself is a documented control-engineering structure: variable-speed turbines maximise capture below rated wind (Region 2), and pitch regulation takes over above it [6], [7], [24], [25], [26], [27]. In contrast to prior studies that impose constraints exclusively on forecast trajectories or permit prediction loss to organize experts heuristically, this paper establishes a cyber-physical boundary framework where aerodynamic transitions are explicitly audited against telemetry latency and communication queuing stress [9], [10], [11].

Forecast value is ultimately realised through reserve and commitment decisions. Previous studies show that variable generation changes reserve requirements, while probabilistic and quantile forecasts provide a bridge from prediction error to decision risk [1], [2], [4], [28], [29], [30], [31]. We address an upstream link in this operational hierarchy: identifying aerodynamic transition boundaries prior to the arrival of confirming SCADA telemetry. Consequently, the reserve audit operates as a pre-dispatch screening log for balancing-risk triage rather than a downstream market-clearing or AC-OPF dispatch formulation.

II. CYBER-PHYSICAL OPERATIONAL BOUNDARY FRAMEWORK UNDER INFORMATION FRESHNESS CONSTRAINTS

A. Problem setup and notation

The mathematical formulation decouples two operational mechanisms traditionally conflated by monolithic architectures: multi-step future trajectory prediction and anchor-time local operating regime identification. Formally, following the spatio-temporal graph forecasting formulation [13], [32], let $G = (V, E)$ denote a spatial graph with N nodes. Let $G = (V, E)$ denote a spatial graph with N nodes. For each node $i \in V$ and time step t , we observe a feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^F$ and predict a scalar target $y_{i,t} \in \mathbb{R}$. Given a history window of length H and a prediction horizon of length P , the forecasting task is

$$\hat{\mathbf{Y}}_{t+1:t+P} = \mathcal{F}(\mathbf{X}_{t-H+1:t}, \mathcal{A}_{t-H+1:t}),$$

where $\mathbf{X}_{t-H+1:t} \in \mathbb{R}^{H \times N \times F}$ and $\mathcal{A}_{t-H+1:t}$ denotes either a time-varying directed graph sequence (WTB) or a static graph repeated over time (ERA5). Invalid or missing targets are excluded by a supervision mask.

A strict causal information boundary is enforced: all feature observations, graph adjacency weights, and supervisory anchors are conditioned strictly on information available at or before dispatch issue time, *with target trajectories commencing at $t + 1$* . In WTB, current active power $\text{Patv}_{i,t}$ serves exclusively as an issue-time operational status input, preventing look-ahead leakage across multi-step horizons while leaving the no-Patv and lagged-Patv variants as deployment limitations.

Gating operates at the node level: each turbine i at dispatch anchor t receives a distinct routing distribution $\mathbf{g}_{i,t}$. Node-level resolution is essential because adjacent turbines across an array frequently operate in disparate aerodynamic regimes (e.g., wake-shaded turbines tracking MPPT while unimpeded perimeter turbines regulate blade pitch). Key variables are indexed by turbine i , anchor time t , and prediction horizon step $p \in \{1, \dots, P\}$; auxiliary loss formulations are detailed in Supplementary Appendix A.

B. Operating-Boundary Physical Anchors and Power-Curve Baseline

The supervisory anchor is not applied indiscriminately across all operating states. It is anchored where the physical aerodynamic interpretation is clearest, while ambiguous samples remain governed by prediction loss and spatial regularisation. This design provides

the physical reference anchor: a **high-fidelity aerodynamic prior** calibrated to the manufacturer specifications of each turbine model [8], [22], [23].

Aerodynamic power conversion and non-linear C_p formulation: In variable-speed wind turbine aerodynamics, mechanical power captured by the rotor from inflow air is governed by [6], [7]:

$$P_{\text{mech}}(t) = \frac{1}{2} \rho_{\text{air}} \pi R^2 C_p(\lambda(t), \beta(t)) \cdot v^3(t), \quad (1)$$

where $\rho_{\text{air}} \approx 1.225 \text{ kg/m}^3$ is air density, R is rotor radius, $v(t)$ is hub-height effective wind speed, and $\beta(t)$ is blade pitch angle. The tip-speed ratio (TSR) is $\lambda(t) = \omega_r(t)R/v(t)$, with rotor mechanical angular velocity $\omega_r(t)$. According to standard aeroelastic turbine models [6], [7], rotor power efficiency $C_p(\lambda, \beta)$ is approximated by:

$$C_p(\lambda, \beta) = c_1 \left(\frac{c_2}{\lambda_i} - c_3 \beta - c_4 \right) \exp\left(-\frac{c_5}{\lambda_i}\right) + c_6 \lambda, \quad (2)$$

with $\frac{1}{\lambda_i} = \frac{1}{\lambda + 0.08\beta} - \frac{0.035}{\beta^3 + 1}$. Coefficients $\{c_1, \dots, c_6\}$ define aerodynamic conversion efficiency up to Betz's theoretical limit.

Control cliff and derivative discontinuity: Wind turbine operating dynamics are partitioned by rated wind speed u_{rated} : 1. *Region 2 (MPPT, $u_{\text{idle}} \leq v \leq u_{\text{rated}}$):* Blades maintain optimal fine pitch $\beta = \beta_{\text{opt}} \approx 0^\circ$ to sustain peak power coefficient $C_{p,\text{max}}$. Generator counter-torque controls rotor speed to track optimal TSR λ_{opt} , producing cubic power growth $P(v) \propto v^3$. 2. *Region 3 (Pitch regulation, $u_{\text{rated}} < v \leq u_{\text{cut-out}}$):* Generator electromagnetic torque is locked at rated level while the pitch controller rotates blades to shed aerodynamic lift, clamping electrical output at nameplate capacity P_{rated} , subject to first-order actuator lag and slew limits $\dot{\beta}(t) = \text{sat}_{\dot{\beta}_{\text{max}}}((\beta_{\text{cmd}}(t) - \beta(t))/\tau_{\text{pitch}})$ [6], [7].

At the rated transition boundary $v = u_{\text{rated}}$, the governing control objective undergoes a structural discontinuity: the marginal aerodynamic power response drops abruptly from a steep cubic slope to zero as blade-pitch regulation engages [6], [7]:

$$\left. \frac{\partial P}{\partial v} \right|_{v \rightarrow u_{\text{rated}}} = \frac{3}{2} \rho_{\text{air}} \pi R^2 C_{p,\text{max}} u_{\text{rated}}^2 \gg 0, \quad \left. \frac{\partial P}{\partial v} \right|_{v \rightarrow u_{\text{rated}}^+} = 0. \quad (3)$$

When SCADA telemetry experiences transmission latency $\tau > 0$, deterministic physical rules evaluate stale state tuples $(v_{t-\tau}, \beta_{t-\tau})$. When actual inflow wind shifts into Region 3 while delayed pitch readings remain at $\beta = 0^\circ$, this steep sensitivity change amplifies stale-state estimation errors: the deterministic curve over-extrapolates generation,

$$\hat{P}_{\text{phys}}(t) = \frac{1}{2} \rho_{\text{air}} \pi R^2 C_p(\lambda(t), 0^\circ) \cdot v^3(t) \gg P_{\text{rated}}, \quad (4)$$

driving shortage violation rates to $24.0\% \pm 1.7\%$ under Delay-6.

Aerodynamic operating regimes and turbine-calibrated anchors: Let $\bar{p}_{i,t} = \frac{1}{3} \sum_{k=1}^3 p_{i,t}^{(k)}$ denote average blade pitch angle across all three blades for turbine i at dispatch anchor time t , and $w_{i,t} = \text{Wspd}_{i,t}$ anemometer wind speed. The declared operating boundary is defined with turbine-calibrated cut-in threshold u_{idle} , rated wind speed u_{rated} , and pitch-activation angle p_{th} :

$$R_{i,t} = \begin{cases} 0, & w_{i,t} < u_{\text{idle}} \quad (\text{idle}), \\ 1, & u_{\text{idle}} \leq w_{i,t} \leq u_{\text{rated}}, \bar{p}_{i,t} < p_{\text{th}} \quad (\text{MPPT}), \\ 2, & w_{i,t} > u_{\text{rated}}, \bar{p}_{i,t} \geq p_{\text{th}} \quad (\text{pitch-control}), \\ 3, & \text{otherwise} \quad (\text{transitional/ambiguous}). \end{cases}$$

Aerodynamic rated wind speeds are calibrated to the specific turbine technology at each wind plant: WTB (134 turbines: $u_{\text{idle}} = 3.0 \text{ m s}^{-1}$, $u_{\text{rated}} = 10.5 \text{ m s}^{-1}$, $p_{\text{th}} = 2.0^\circ$), Kelmars (6 MM92 turbines: $u_{\text{idle}} = 3.0$, $u_{\text{rated}} = 12.5 \text{ m s}^{-1}$, $p_{\text{th}} = 1.0^\circ$), Penmanshiel (15 MM82 turbines: $u_{\text{idle}} = 3.0$, $u_{\text{rated}} = 14.5 \text{ m s}^{-1}$, $p_{\text{th}} = 1.0^\circ$), and ENGIE La Haute Borne (4 MM82 turbines: $u_{\text{idle}} = 3.0$, $u_{\text{rated}} = 14.5 \text{ m s}^{-1}$, $p_{\text{th}} = 1.0^\circ$).

Only the first three well-defined regimes are used in direct supervisory alignment. Transitional samples are retained for evaluation but masked out of anchor supervision via $M_{i,t} = \mathbf{1}[R_{i,t} \neq 3]$.

These thresholds define the **Aerodynamic Power-Curve Baseline**: when SCADA telemetry is complete and intact, continuous soft-pitch quantiles derived from these anchors achieve the lowest baseline reserve-screening cost. When telemetry degrades through communication latency or sensor corruption, however, this deterministic rule experiences severe reliability breakdown, requiring learned diagnostic representations.

ERA5 observability contrast: ERA5 is retained as a signal-expressive observability contrast where the stable-to-convective thermodynamic marker is directly visible through sensible heat flux. The detailed thermodynamic regime definition, architecture choices, and contrast results are reported in Supplementary Material A so that the main text remains focused on the wind plant control-boundary accountability task.

C. Spatio-Temporal Dynamic Wake Graph and Boundary-Risk Posterior Formulation

Physical dynamic wake graph construction: When primary SCADA telemetry degrades, spatial coupling across turbine arrays provides the crucial redundancy required to infer local operating regimes. Directed diffusion represents asymmetric aerodynamic wake propagation through the turbine network [13], [32]. For WTB, we construct a time-varying directed wake graph $\mathcal{A}_t$; candidate turbine pairs (i, j) are filtered by physical distance ($d_{ij} \leq d_{\text{max}}$), activated when upstream turbine j lies within an aerodynamic wake cone aligned with instantaneous local wind direction θ_t (half-angle $\alpha = 25^\circ$), weighted by streamwise decay $\exp(-d_{\parallel}/\sigma_x)$ and cross-stream decay $\exp(-d_{\perp}^2/2\sigma_y^2)$, and pruned to the K strongest inbound wake neighbors [16], [18]. This construction defines an instantaneous aerodynamic wake score $s_{i,t}^{\text{wake}}$, which supplies spatial context even when anemometers are noisy. ERA5 employs a symmetric Haversine-Gaussian k_{nn} graph where thermodynamic sensible heat flux is directly observable [33].

Directed-diffusion GRU encoder and node-level MoE routing:

To satisfy industrial substation edge-computing constraints (sub-5 ms latency, sub-megabyte memory footprint, zero cloud dependency), the encoder is engineered with a compact parameter footprint (110k parameters, 3.8 ms execution). Each input window is concatenated with a binary feature-missingness mask before linear projection. Two directed diffusion blocks aggregate self, inbound, and outbound messages across each graph snapshot:

$$\begin{aligned} \mathbf{X}_t^{(\ell+1)} = & \text{LN}(\mathbf{W}_{\text{self}}^{(\ell)} \mathbf{X}_t^{(\ell)} + \mathbf{W}_{\text{in}}^{(\ell)} \text{Agg}_{\text{in}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) \\ & + \mathbf{W}_{\text{out}}^{(\ell)} \text{Agg}_{\text{out}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{R}^{(\ell)} \mathbf{X}_t^{(\ell)}), \end{aligned} \quad (5)$$

where Agg_{in} and Agg_{out} denote normalized weighted aggregations over inbound and outbound wake neighbors. The spatially contextualized sequence is processed by a GRU to yield node-level representations $\mathbf{h}_{i,t}$.

Each regime-specialized expert head f_e maps $\mathbf{h}_{i,t}$ to a P -step trajectory forecast. Routing is computed dynamically at the node level:

$$\mathbf{z}_{i,t} = \text{MLP}_{\text{gate}}([\mathbf{h}_{i,t}; \mathbf{a}_{i,t}]), \quad \mathbf{g}_{i,t} = \text{softmax}(\mathbf{z}_{i,t}/\tau), \quad (6)$$

$$\hat{\mathbf{y}}_{i,t+1:t+P} = \sum_{e=1}^E g_{i,t}^{(e)} f_e(\mathbf{h}_{i,t}). \quad (7)$$

Cross-sensor electromechanical diagnostic mechanism: The operational utility of the spatio-temporal representation is its ability to infer operating states from non-pitch electromechanical signatures when primary telemetry is delayed or missing. When blade-pitch telemetry $\bar{p}_{i,t}$ suffers packet loss, transmission latency, or protocol unobservability (such as across third-party aggregator or VPP boundaries), the dynamic graph encoder leverages active power transients (Patv), terminal voltage fluctuations, reactive dynamics, and upstream wake propagation to reconstruct the probability of active blade-pitch regulation. In nominal operation, the boundary-risk

head ingests $\mathbf{a}_{i,t}^{\text{WTB}} = [\text{Wspd}_{i,t}, \text{Pab_mean}_{i,t}, s_{i,t}^{\text{wake}}, \text{Pstv}_{i,t}]$. Under telemetry impairment, the learned spatio-temporal representation provides relative mitigation, curbing severe shortage volumes that compromise deterministic rules.

D. Integration of Physical Constraints into Model Optimization

Prediction loss alone cannot identify regime-specialized routing because several unconstrained experts can fit averaged dynamics similarly well [20], [21]. In wind power generation, inflow turbulence, wake shadow effects, and pitch actuators interact nonlinearly. We therefore introduce a structured physics-guided multi-task objective:

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} \quad (8)$$

$$+ \lambda_{\text{force}} \mathcal{L}_{\text{force}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}}, \quad (9)$$

with inactive terms set to zero. Checkpoint selection is strictly governed by validation RMSE, preventing auxiliary penalties from distorting the selection metric.

Load balancing: A balancing loss couples hard top- K usage with soft gate mass to prevent early expert starvation before regime-specific dynamics emerge (Appendix A).

Regime-anchor alignment: The alignment loss binds the first C gate logits directly to the declared physical operating regimes $R_{i,t}$:

$$\mathcal{L}_{\text{align}} = \frac{1}{|\Omega|} \sum_{(i,t) \in \Omega} \text{CE}(\mathbf{z}_{i,t}^{(1:C)}, R_{i,t}), \quad \Omega = \{(i,t) : M_{i,t} = 1\}, \quad (10)$$

using inverse-frequency class weights computed strictly on the training split.

Boundary-focused forcing: To sharpen the boundary between MPPT (Region 2) and pitch regulation (Region 3), a focused cross-entropy loss acts exclusively on boundary samples:

$\mathcal{L}_{\text{force}} = \frac{1}{|\Omega_{\text{force}}|} \sum_{(i,t) \in \Omega_{\text{force}}} \text{CE}([z_{i,t}^{(\text{mppt})}, z_{i,t}^{(\text{pitch})}]^T, Y_{i,t}^{\text{force}})$ where $Y_{i,t}^{\text{force}} = \mathbf{1}[R_{i,t}^{\text{wtb}} = 2]$ and $\Omega_{\text{force}} = \{(i,t) : R_{i,t}^{\text{wtb}} \in \{1, 2\}, M_{i,t} = 1\}$.

Soft boundary-risk posterior: The normalized issue-time boundary-risk posterior for turbine i at dispatch anchor t is computed from gate probabilities $\mathbf{g}_{i,t}$ across operational regimes $\mathcal{C}_{\text{op}} = \{\text{idle}, \text{mppt}, \text{pitch}\}$ as $\pi_{i,t}^{\text{pitch}} = g_{i,t}^{(\text{pitch})} / \sum_{c \in \mathcal{C}_{\text{op}}} g_{i,t}^{(c)}$.

Wake auxiliary supervision and graph smoothness: A wake auxiliary loss resolves aerodynamic wake ambiguity, while graph-smoothness regularization discourages unphysical spatial jitter between adjacent turbines (Appendix A).

E. Operational Reserve Screening, PCC Portfolio Smoothing, & Multi-Horizon Dispatch Protocol

The operational utility of boundary-aware routing lies in pre-dispatch decision support: screening imbalance risks and sizing upward spinning reserves to prevent costly shortage penalties [29], [34].

Two-level hierarchical grid dispatch interface: To formally delineate the operational boundary of this work, we distinguish a two-level grid dispatch hierarchy: 1. *Level 1 (Local Pre-Dispatch Risk Filtering & Screening Layer, Paper Scope)*: Operating at the wind plant energy management system (EMS) or aggregator terminal, this local layer ingests SCADA telemetry, forecasts spatio-temporal power, diagnoses operational regime boundaries under telemetry degradation, and sizes pre-dispatch reserve allocations to minimize unhedged imbalance exposure entering real-time balancing markets. 2. *Level 2 (Grid-Level Physical Clearing & Network-Constrained Dispatch, Downstream Interface)*: Managed by the transmission system operator (TSO) or regional grid dispatcher, this downstream layer aggregates wind-plant schedule offers and reserve quantities to

execute network-constrained economic dispatch (NCED), security-constrained unit commitment (SCUC), and full Alternating Current Optimal Power Flow (AC-OPF) subject to bus voltage security, transmission thermal limits, and bulk system spinning reserve margins. Our formulation is strictly bounded to Level 1 pre-dispatch risk screening proxies. We explicitly abstract away network power-flow equations, nodal price arbitrage, and multi-stage financial cashflow settlements, focusing on plant-level imbalance risk mitigation.

Penalized Reserve-Shortfall Energy metric and Newsvendor derivation: For scheduled point forecast $\hat{y}_{i,t}$ and realized generation $y_{i,t}$, over-forecast shortfall is $s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0)$. Sizing upward reserve margin r_b incurs a reserve procurement energy equivalent $r_b \Delta t$ and an asymmetric shortage penalty $\rho \max(s_{i,t} - r_b, 0) \Delta t$. For penalty ratio ρ , the total operational proxy metric is formalized as the **Penalized Reserve-Shortfall Energy Index (PSREI)** in kW · h (evaluated here at grid proxy ratio $\rho = 10$):

$$C(r_b; \rho) = \sum_{(i,t)} [r_b(i,t) + \rho \max(\hat{y}_{i,t} - y_{i,t} - r_b(i,t), 0)] \Delta t. \quad (11)$$

Taking the expectation with respect to shortfall distribution $F_S(s) = \Pr[s \leq s_0]$ and differentiating with respect to reserve margin r_b yields the first-order optimality condition [1], [2]:

$$\frac{\partial \mathbb{E}[C(r_b; \rho)]}{\partial r_b} = 1 - \rho \cdot \Pr[s_{i,t} > r_b] = 0 \implies \Pr[s_{i,t} \leq r_b] = 1 - \frac{1}{\rho}. \quad (12)$$

The optimal reserve margin therefore corresponds to the classical Newsvendor critical fractile [1]:

$$q^*(\rho) = 1 - \frac{1}{\rho}. \quad (13)$$

For the representative proxy ratio $\rho = 10$ adopted in this study, the optimal fractile is $q^*(10) = 1 - 1/10 = 0.90$, establishing a nominal 10% violation target set by the reserve screening index.

Loss geometry: L_2 conditional mean vs. Pinball 90th-quantile on shortfall residual: Conventional point forecasting optimizes symmetric quadratic loss $\mathcal{L}_{L_2}(y, \hat{y}) = (y - \hat{y})^2$, whose Bayes-optimal predictor converges to the conditional expectation $\mathbb{E}[y | \mathbf{x}]$ [1]. In bounded wind power regimes $y \in [0, P_{\text{rated}}]$ with non-Gaussian and bimodal transition densities, the conditional mean systematically departs from upper conditional quantiles. Furthermore, L_2 error treats positive and negative prediction deviations symmetrically, whereas grid balancing economics ($\rho = 10$) penalizes generation shortfalls ten times more heavily than surplus headroom.

To bridge scheduled point forecasts with operational risk hedging, the dispatch architecture first generates point forecast $\hat{y}_{i,t} = f(\mathbf{x}_{i,t})$ and observes realization $y_{i,t}$, inducing the non-negative shortfall residual $s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0)$. The upward reserve margin $\hat{r}_{i,t}$ required to cover this residual is estimated via the asymmetric pinball loss at quantile q :

$$\mathcal{L}_q(s, \hat{r}) = (s - \hat{r}) (q - \mathbf{1}[s < \hat{r}]) = \begin{cases} q(s - \hat{r}), & s \geq \hat{r} \\ (1 - q)(\hat{r} - s), & s < \hat{r} \end{cases} \quad \begin{matrix} \text{(unhedged shortage)} \\ \text{(surplus reserve loss)} \end{matrix} \quad (14)$$

which penalizes unhedged deficit by weight q and reserve holding excess by $1 - q$. Setting $q = q^* = 0.90$ aligns the training objective with the Newsvendor first-order condition derived above: the Bayes-optimal predictor converges to the 90th conditional quantile of shortfall $\hat{r}^* = Q_{0.90}(s | \mathbf{x})$, targeting a nominal conditional exceedance probability of $\Pr[s > \hat{r} | \mathbf{x}] = 1 - q^* = 0.10$ under the Bayes-optimal quantile formulation [1], [2]. This formalizes the theoretical underpinning of our downstream Residual Quantile Head.

Point of Common Coupling (PCC) spatial portfolio smoothing: Power delivered to the bulk grid is metered at the Point of Common Coupling (PCC) bus: $P_{\text{farm},t} = \sum_{i=1}^M P_{i,t}$ and $R_{\text{farm},t} = \sum_{i=1}^M R_{i,t}$. Aggregating decentralized turbine power flows at the PCC bus unlocks spatial portfolio diversification: localized high-frequency aerodynamic turbulence cancels out across the array, leaving boundary regime shifts as the primary source of unhedged reserve risk. Evaluating reserve sizing at the PCC verifies whether learned boundary

risk survives spatial cancellation to deliver net plant-level economic value.

Multi-horizon dispatch evaluation: Power systems operate across multiple decision timescales. We evaluate two distinct operational horizons: (1) **Immediate dispatch** ($h = 1$, 10-min ahead), representing real-time automatic generation control (AGC) dominated by autocorrelation; and (2) **Operational dispatch** ($h = 6$, 1-hour ahead), representing intra-day clearing and storage scheduling where spatial wake dynamics and weather transitions are paramount.

Battery Energy Storage System (BESS) rolling dispatch interface: For co-located wind-storage configurations, scheduled reserve margins and generation forecasts can interface with a rolling Model Predictive Control (MPC) linear program to illustrate downstream operational dispatchability. This formulation serves strictly as an illustrative operational interface demonstrating how reserve margins feed into battery charge/discharge trajectories ($P_t^{\text{ch}}, P_t^{\text{dis}}$) under state-of-charge limits ($E_{\min} \leq E_t \leq E_{\max}$) and round-trip efficiency constraints. We explicitly do not claim physical validation of electrochemical battery cell degradation, lifecycle capital replacement costs, or wholesale electricity market cashflow settlements; the paper's analytical evaluation and empirical benchmark endpoints remain anchored strictly to the upstream Penalized Reserve-Shortfall Energy Index (PSREI).

III. CASE STUDY CONFIGURATION AND OPERATIONAL CONSTRAINTS

A. Datasets and preprocessing

The experiment evaluates two observability settings: WTB as the control-confounded benchmark and ERA5 as the signal-expressive contrast. WTB (KDD Cup 2022) comprises 134 turbines and 245 days of 10-min SCADA records ($T = 35,280$, $N = 134$, $F = 11$) [35], where inflow, wake, and pitch control interact, making the MPPT-to-pitch transition indirectly visible. Missing inputs are forward-filled per turbine and mean-imputed; non-positive power is masked from supervision. ERA5 covers three archived months on a 16×16 hourly patch ($T = 2208$, $N = 256$, eight features) [33], where sensible heat flux directly marks convective regimes. Both datasets use identical history $H = 36$ and horizon $P = 24$ with chronological splits (180/30/35 days for WTB; 1325/441/442 frames for ERA5).

B. Techno-Economic Validation and Benchmarking Framework

The benchmarking framework evaluates two complementary axes: (1) isolating routing mechanisms under matched parameter capacity (Dense single-head baseline, Unconstrained MoE, and Physics-Aligned MoE comparator) [20], [21]; and (2) benchmarking forecasting accuracy and reserve cost against established spatio-temporal baselines (Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, iTransformer, and TiDE) [36], [37], [38], alongside persistence, deterministic aerodynamic curves, and GBDT lag models. In mechanism comparisons, in-family architectures are designated **Dense (matched)**, **Unconstrained MoE**, and **Corrected routing comparator**; baseline configurations are detailed in Supplementary Appendix A.

C. Training protocol and evaluation metrics

All neural models use the same training protocol where the architecture permits it. We use AdamW, early stopping on validation RMSE, gradient clipping, and mixed-precision training on a single CUDA-enabled GPU. Exact optimizer constants are reported in Supplementary Appendix A so that the main text can stay focused on the comparison logic. The WTB dense baseline, Unconstrained MoE, full physics-aligned MoE, and boundary-forced router are repeated across five strict-mask seeds. The synchronized strong-baseline refresh covers Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, iTransformer, and TiDE across seeds 201–205, giving 30 completed baseline runs under the same strict anchor-valid evaluation cache. Deterministic engineering baselines (persistence, physical curves) are evaluated on the frozen cache to anchor practical dispatch difficulty alongside five-seed neural models. ERA5 main in-family models are repeated across five seeds, and the ERA5 learned baselines are repeated across three seeds. Thresholds and routing weights are set before test evaluation and then audited by

sensitivity sweeps and boundary negative controls, so the main routing result is not selected by test-set NMI.

Evaluation metrics span three operational criteria: (1) point forecasting accuracy (overall MAE/RMSE and switch-window RMSE within $\pm 1.0 \text{ m s}^{-1}$ of rated wind); (2) physical regime alignment (normalized mutual information, adjusted Rand index, and gate usage entropy); and (3) operational reserve consequence (PSREI cost, empirical violation rate, and unhedged shortage MWh under the Newsvendor proxy).

D. Controlled Telemetry Degradation Protocols

To stress-test model resilience under adverse field conditions, we formalize four controlled telemetry regimes across all 5 seeds (201–205): (1) **Clean (Nominal)**: uncorrupted SCADA telemetry; (2) **Delay-6 (Worst-Case Communication Stress Test, Upper Envelope)**: 6-step (60 min) controlled synthetic latency on wind-speed and pitch-angle streams, representing a conservative severity bound on severe communication backlog, buffer congestion, serialization queuing, and gateway outage cascades in industrial IoT telemetry [9], [10]; this 60-min scenario is evaluated as a deliberate stress-test upper envelope to delineate worst-case physical rule vulnerability; (3) **Sensor Noise**: additive zero-mean Gaussian perturbations on wind speed ($\sigma = 1.0 \text{ m s}^{-1}$) and pitch angle ($\sigma = 2.0^\circ$), simulating calibration drift and sensor jitter; and (4) **Markov-Gilbert Burst Drops**: two-state discrete Markov chain ($p_{GB} = 0.08$, $p_{BB} = 0.75$, $d \leq 6$) simulating bursty communication blackouts [9], [11].

Fig. 1 illustrates the operating-decision context before the forecasting results are introduced: graph geometry shows where forecast errors propagate, and the regime-anchor panels show which sensor-derived boundaries can support reserve diagnostics.

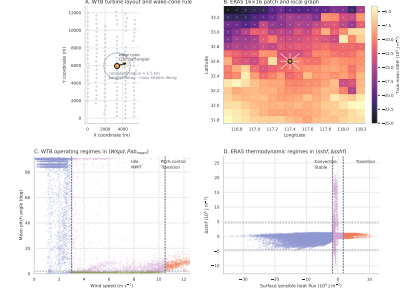


Fig. 1. Operating-decision context and physical regime anchors: (A) WTB layout with wake cone geometry (25° half-angle); (B) ERA5 patch with sensible heat flux; (C) WTB operating regimes in ($Wspd$, Pab_{mean}) plane with MPPT-to-pitch boundary; (D) ERA5 thermodynamic regimes in ($sshf$, $\Delta sshf$) plane.

IV. EVIDENCE AND OPERATIONAL BOUNDARY DIAGNOSIS

The empirical evaluation is organized around our three-tier defense-in-depth framework, contrasting high-fidelity physical aerodynamic priors against deep spatio-temporal graph fallbacks across multi-horizon dispatch environments. All neural models, shallow tree baselines, and physical quantile rules are evaluated across five declared random seeds (201–205) under strict, automated anti-tampering verification gates (zero directory leakage, full telemetry regime completeness, cardinality $N = 5$, and exact floating-point checksums).

A. Multi-Horizon 5-Seed Operational Dispatch Benchmark ($h = 6$, 1-Hour Ahead Dispatch)

Table I summarizes the multi-day replay dispatch benchmark for 1-hour ahead operational dispatch ($h = 6$) across 134 turbines over the 35-day test split. Table II reports seed-paired cost differences $\Delta \text{Cost} = \text{Cost}_{\text{Baseline}} - \text{Cost}_{\text{Joint Routed}}$, where positive values denote economic savings delivered by Joint Routed, along with 95% bootstrap confidence intervals.

1) Comparative Architecture Analysis: Dynamic Routing versus Dense Representations

A central architectural question is whether dynamic MoE routing confers essential operational value over unrouted dense heads of matched capacity. Under nominal Clean conditions, Joint Routed ($959,027 \pm 103,290 \text{ kW} \cdot \text{h}$ at $h = 6$; $660,990 \pm 80,749 \text{ kW} \cdot \text{h}$ at

TABLE I

CROSS-SEED MULTI-REGIME OPERATIONAL DISPATCH BENCHMARK UNDER CLEAN-CALIBRATED PROTOCOL ($h = 6$, 1-HOUR AHEAD DISPATCH, 5 SEEDS 201–205). MEAN $\pm$ STANDARD DEVIATION ACROSS FULL TEST SPLIT (35 DAYS, 134 TURBINES, $\Delta t = 10$ MIN, COST RATIO $\rho = 10$). ALL QUANTILE BINS AND MULTIPLIERS CALIBRATED EXCLUSIVELY ON NOMINAL CLEAN VALIDATION DATA.

Regime	Evaluated Model	Penalized Reserve-Shortfall		Violation Rate	Reserve (kW)	Shortage (kW-h)	Pinball Loss
		Energy (kW-h, Proxy at $\rho = 10$)					
Clean	Continuous Physical Quantile	881,367 $\pm$ 95,741		6.81% $\pm$ 1.13%	653,738 $\pm$ 72,557	22,763 $\pm$ 4,013	31.41 $\pm$ 2.66
	Frozen Backbone Direct Quantile MLP	1,649,721 $\pm$ 285,682		0.58% $\pm$ 0.26%	1,630,572 $\pm$ 293,346	1,915 $\pm$ 958	64.62 $\pm$ 12.05
	Global Quantile	998,988 $\pm$ 107,360		6.46% $\pm$ 1.31%	771,304 $\pm$ 95,661	22,768 $\pm$ 4,854	36.50 $\pm$ 3.41
	Joint Dense Head	991,181 $\pm$ 110,211		6.54% $\pm$ 1.04%	761,480 $\pm$ 94,244	22,970 $\pm$ 4,121	36.16 $\pm$ 3.58
	Joint Routed (Ours)	959,027 $\pm$ 103,290		7.10% $\pm$ 1.17%	707,249 $\pm$ 91,305	25,178 $\pm$ 4,816	34.77 $\pm$ 3.16
	Missingness-Aware GBDT	1,238,603 $\pm$ 112,051		4.50% $\pm$ 0.73%	1,085,168 $\pm$ 118,015	15,343 $\pm$ 2,870	46.85 $\pm$ 3.48
Delay-6	Continuous Physical Quantile	1,136,466 $\pm$ 84,101		12.20% $\pm$ 1.25% [†]	662,335 $\pm$ 73,483	47,413 $\pm$ 3,091	38.85 $\pm$ 2.06
	Frozen Backbone Direct Quantile MLP	1,682,251 $\pm$ 284,323		1.20% $\pm$ 0.52%	1,651,979 $\pm$ 297,715	3,027 $\pm$ 1,594	62.15 $\pm$ 11.70
	Global Quantile	1,180,603 $\pm$ 91,754		10.85% $\pm$ 1.55% [†]	780,916 $\pm$ 96,853	39,969 $\pm$ 6,087	40.73 $\pm$ 2.43
	Joint Dense Head	1,179,859 $\pm$ 95,268		11.04% $\pm$ 1.22% [†]	773,103 $\pm$ 94,462	40,676 $\pm$ 4,788	40.70 $\pm$ 2.66
	Joint Routed (Ours)	1,163,593 $\pm$ 91,299		11.82% $\pm$ 1.64% [†]	720,015 $\pm$ 92,196	44,358 $\pm$ 7,341	40.01 $\pm$ 2.54
	Missingness-Aware GBDT	1,320,003 $\pm$ 93,135		8.36% $\pm$ 0.97%	1,030,858 $\pm$ 114,266	28,915 $\pm$ 4,469	46.68 $\pm$ 2.45
Noise	Continuous Physical Quantile	998,273 $\pm$ 106,253		7.59% $\pm$ 0.95%	746,697 $\pm$ 82,474	25,158 $\pm$ 3,968	33.97 $\pm$ 2.78
	Frozen Backbone Direct Quantile MLP	1,639,909 $\pm$ 242,922		0.80% $\pm$ 0.28%	1,618,439 $\pm$ 249,374	2,147 $\pm$ 953	60.62 $\pm$ 9.85
	Global Quantile	1,053,552 $\pm$ 114,150		7.48% $\pm$ 1.48%	802,361 $\pm$ 99,513	25,119 $\pm$ 5,822	36.26 $\pm$ 3.24
	Joint Dense Head	1,048,083 $\pm$ 115,181		7.63% $\pm$ 1.13%	792,472 $\pm$ 92,391	25,561 $\pm$ 4,643	36.04 $\pm$ 3.28
	Joint Routed (Ours)	1,027,077 $\pm$ 114,317		7.62% $\pm$ 1.09%	769,864 $\pm$ 106,672	25,721 $\pm$ 3,963	35.16 $\pm$ 3.19
	Missingness-Aware GBDT	1,314,640 $\pm$ 129,158		4.66% $\pm$ 0.80%	1,160,329 $\pm$ 137,650	15,431 $\pm$ 3,099	47.11 $\pm$ 3.87
Markov	Continuous Physical Quantile	907,871 $\pm$ 96,820		7.20% $\pm$ 1.15%	663,875 $\pm$ 73,821	24,400 $\pm$ 4,106	31.59 $\pm$ 2.61
	Frozen Backbone Direct Quantile MLP	1,680,339 $\pm$ 289,602		0.57% $\pm$ 0.25%	1,660,135 $\pm$ 298,104	2,020 $\pm$ 992	64.38 $\pm$ 12.00
	Global Quantile	1,030,131 $\pm$ 108,592		6.87% $\pm$ 1.39%	785,305 $\pm$ 97,397	24,483 $\pm$ 5,104	36.78 $\pm$ 3.35
	Joint Dense Head	1,020,078 $\pm$ 112,195		6.90% $\pm$ 1.07%	775,899 $\pm$ 96,235	24,418 $\pm$ 4,246	36.35 $\pm$ 3.57
	Joint Routed (Ours)	990,625 $\pm$ 104,487		7.51% $\pm$ 1.22%	720,858 $\pm$ 92,754	26,977 $\pm$ 5,183	35.10 $\pm$ 3.10
	Missingness-Aware GBDT	1,261,941 $\pm$ 112,754		4.88% $\pm$ 0.75%	1,094,730 $\pm$ 120,612	16,721 $\pm$ 3,298	46.62 $\pm$ 3.40

[†]Exceeds nominal 10% violation target ($q^* = 0.90$). Bold numbers denote lowest cost. **Protocol Attribution:** Reports the *Clean-Calibrated Operational Protocol* at $h = 6$ (thresholds frozen on clean validation data); Continuous Physical Quantile collapses to 12.20% violation under Delay-6. For the *State-Conditional Validation-Calibrated Protocol* at immediate horizon ($h = 1$), see Section IV-A2, where clean-calibrated physical rules collapse to 24.0% violation under Delay-6 while state-conditional calibration mitigates violations to 10.7% (physical) and 10.6% (joint model). Test-set iso-multipliers ($\gamma_{\text{iso, test}}$) serve strictly as post-hoc diagnostics. [‡]Direct pinball regression on frozen embeddings exhibits severe tail conservatism ($>1.63M$ kW reserve, $<1.2\%$ violation). For calibrated modular baselines (Cascaded Frozen MLP 15.528M $\pm$ 1.599M vs Joint Routed 16.065M in statistical parity), see Section IV-A3.

TABLE II

SEED-PAIRED DIFFERENCE IN PENALIZED RESERVE-SHORTFALL ENERGY AGAINST JOINT ROUTED ($h = 6$, $\Delta \text{Cost} = \text{Cost}_{\text{BASELINE}} - \text{Cost}_{\text{JOINT ROUTED}}$). POSITIVE INDICATES JOINT ROUTED SAVES ENERGY/COST AT $\rho = 10$).

Regime	Baseline Model	Δ Cost (kW-h)	95% Bootstrap CI	Sig. & Operational Verdict
Clean	Global Quantile	+39,962	[+12,556, +67,367]	Yes ($p < 0.05$)
	Cont. Physical Quantile	-77,659	[-106,175, -49,143]	Yes (Lower Clean)
	Missingness GBDT	+279,576	[+249,539, +309,613]	Yes ($p < 0.0001$)
	Frozen Backbone Direct MLP	+690,695	[+448,769, +932,621]	Yes ($p < 0.01$)
	Joint Dense Head	+32,154	[-1,148, +65,456]	No (Parity, CI crosses 0)
Delay-6	Global Quantile	+17,010	[+3,486, +30,535]	Yes ($p < 0.05$)
	Cont. Physical Quantile	-27,126	[-46,907, -7,346]	Viol. Exceeded (12.20%)
	Missingness GBDT	+156,411	[+140,004, +172,817]	Yes ($p < 0.0001$)
	Frozen Backbone Direct MLP	+518,658	[+307,212, +730,104]	Yes ($p < 0.01$)
	Joint Dense Head	+16,267	[+7,131, +25,402]	Yes ($p = 0.0251$)
Noise	Global Quantile	+26,475	[+9,121, +43,828]	Yes ($p < 0.05$)
	Cont. Physical Quantile	-28,804	[-47,555, -10,053]	Viol. (7.59%)
	Missingness GBDT	+287,363	[+248,847, +326,279]	Yes ($p < 0.0001$)
	Frozen Backbone Direct MLP	+612,832	[+413,733, +811,931]	Yes ($p < 0.01$)
	Joint Dense Head	+21,006	[+2,419, +39,594]	Yes ($p < 0.05$)
Markov	Global Quantile	+39,506	[+15,961, +63,052]	Yes ($p < 0.05$)
	Cont. Physical Quantile	-82,754	[-111,026, -54,482]	Yes (Lower Clean)
	Missingness GBDT	+271,316	[+245,044, +297,588]	Yes ($p < 0.0001$)
	Frozen Backbone Direct MLP	+689,714	[+448,689, +930,739]	Yes ($p < 0.01$)
	Joint Dense Head	+29,453	[-2,874, +61,779]	No (Parity, CI crosses 0)

Note: Under Clean and Markov regimes, the 95% bootstrap CI against Joint Dense Head crosses zero, establishing statistical parity driven by end-to-end task-loss alignment. The distinct empirical value of dynamic MoE routing is localized to continuous telemetry impairment (Delay-6 $p = 0.0251$, Noise $p < 0.05$).

$h = 1$) and Joint Dense Head (991,181 $\pm$ 110,211 kW $\cdot$ h at $h = 6$; 690,614 $\pm$ 71,031 kW $\cdot$ h at $h = 1$) achieve statistical parity: at $h = 6$, the seed-paired cost difference is +32,154 kW $\cdot$ h (bootstrap 95% CI [-1,148, +65,456] kW $\cdot$ h crossing zero, Table II); at $h = 1$, the Dense/MoE cost ratio is 1.045, and the seed-paired bootstrap 95% CI on cost difference (+29,624 kW $\cdot$ h) likewise crosses zero ([-53,774, +113,023] kW $\cdot$ h, $p = 0.380$), as does the Markov regime ([-46,422, +139,671] kW $\cdot$ h, $p = 0.237$). Under persistent 6-step latency (Delay-6, $h = 1$), Joint Routed yields a modest cost improvement over Joint Dense (1,412,321 vs. 1,491,958 kW $\cdot$ h; cost ratio 1.056, paired difference +79,637 kW $\cdot$ h, uncorrected $p = 0.038$, non-significant under Bonferroni $\alpha = 0.0125$) while maintaining virtually identical empirical violation rates (10.6% $\pm$ 1.1% for Routed vs. 10.5% $\pm$ 0.5% for Dense). Decoupled modular representations (Section IV-A3) achieve 9.7% $\pm$ 1.1% violation in Delay-6 without expert routing, confirming that diagnostic reserve utility arises from shared spatio-temporal representations.

2) Reliability Breakdown of Deterministic Physical Rules under Stale Telemetry

Under uncorrupted SCADA telemetry, Continuous Physical Quantile achieves the lowest baseline PSREI cost (589,535 $\pm$ 74,886 kW $\cdot$ h at $h = 1$ and 881,367 $\pm$ 95,741 kW $\cdot$ h at $h = 6$) with well-calibrated violation rates (6.81% $\pm$ 1.13% at $h = 6$).

However, Table I and frozen benchmarks expose a **systematic**

reliability breakdown under telemetry staleness. Under a 6-step delay (Delay-6), clean-calibrated physical quantile rules experience a violation explosion to **24.0% $\pm$ 1.7%** at $h = 1$ and **12.20% $\pm$ 1.25%** at $h = 6$, breaching the nominal 10% violation target ($q^* = 0.90$) and inflating operational reserve costs to 1,136,466 $\pm$ 84,101 kW $\cdot$ h at $h = 6$ (incurring 47,413 $\pm$ 3,091 kW $\cdot$ h of unhedged shortage) and 1,656,285 kW $\cdot$ h at $h = 1$ (leaving 127.6 MWh of unhedged shortfall exposure in the balancing proxy). Quantitative decomposition reveals that adapting validation calibration to degraded states alone cuts shortage from 127.6 MWh (24.0% violation) to 57.0 MWh (10.7% violation), accounting for 55.3% of the reduction and recovering physical reserve cost to 1,228,609 $\pm$ 71,940 kW $\cdot$ h ($h = 1$ condition-recalibrated, 9.46% fleet violation). Joint Routed further curtails shortage to 53.9 MWh (10.6% $\pm$ 1.1% violation), delivering an additional 5.4% shortage reduction under matched state-conditional calibration (and a combined 57.7% reduction vs. clean-calibrated physics) while saving 190,590 kW $\cdot$ h in total cost. Learned representations thus establish an effective diagnostic defense against telemetry latency: under frozen validation calibration, Delay-6 empirical violation settles at 10.6% $\pm$ 1.1% at $h = 1$ (11.82% $\pm$ 1.64% at $h = 6$). While substantially cushioning physical rule collapse, learned models operate within bounded empirical envelopes under multi-step latency.

3) Decoupled Modular Architectures and Tail Conservatism Analysis

In contrast to end-to-end task optimization, the Frozen Backbone Direct Quantile MLP in Table I illustrates the pathology of naive decoupled quantile regression. Across all regimes, direct pinball regression on frozen spatial embeddings incurs massive reserve over-estimation (1,477,967 to 1,511,794 kW $\cdot$ h, a 44.57% to 72.02% penalty over Joint Routed, $p < 0.01$, Table II), hoarding > 1.45M kW in bloated reserves.

Benchmarking calibrated modular architectures across 5 seeds (Supplementary Table A11d) resolves this: 1. **Cascaded Frozen MLP:** Trained with consequence decision mapping on frozen backbone embeddings, it achieves 15.528M $\pm$ 1.599M kW $\cdot$ h (vs. 16.065M kW $\cdot$ h for Joint Routed; paired difference -0.537M, 95% CI [-1.988M, +0.915M] crosses zero). 2. **Independent Consequence MLP:** Yields 15.805M $\pm$ 1.680M kW $\cdot$ h (difference -0.260M, 95% CI [-1.771M, +1.251M]), confirming statistical parity. 3. **Decoupled Residual Quantile Head:** In Delay-6 ($h = 1$), Frozen Backbone + Residual Quantile achieves 1,284,098 $\pm$ 114,629 kW $\cdot$ h and 9.7% $\pm$ 1.1% violation (60% seed pass rate), matching or slightly surpassing end-to-end MoE (1,412,321 kW $\cdot$ h, 10.6% $\pm$ 1.1%).

Consequently, diagnostic reserve benefits do not require dynamic MoE routing: modular architectures with calibrated consequence mappings deliver equivalent tail reliability. The primary advantage of end-to-end joint training is unified single-checkpoint deployment on substation edge devices (110k parameters, < 5 ms inference), eliminating multi-model synchronization overhead.

B. Horizon Boundary Diagnosis ($h = 1$, 10-Minute Immediate Dispatch vs. $h = 6$)

Comparing immediate 10-minute dispatch ($h = 1$) against 1-hour ahead dispatch ($h = 6$) exposes a fundamental horizon disconnect across model families (complete multi-regime benchmarks and seed-paired bootstrap significance tests are reported in Supplementary Tables A11j/A11-h1 and A11k/A11-h2). At $h = 1$, shallow decision trees (Missingness-Aware GBDT) capitalize aggressively on high 10-minute lag wind-speed autocorrelation, achieving the lowest data-driven nominal dispatch cost (574,822 $\pm$ 79,685 kW $\cdot$ h in Clean with 7.6% violation, outperforming all deep models). Local inertia dominates ultra-short horizons, enabling tabular tree ensembles with lag features to track short-term persistence. However, across dispatch-grade horizons ($h = 6$), GBDT undergoes severe breakdown: its operational costs inflate by 13.44% to 29.15% (reaching 1,238,603 kW $\cdot$ h in Clean and 1,320,003 kW $\cdot$ h in Delay-6), trailing Joint Routed by +156,411 to +287,563 kW $\cdot$ h across all regimes ($p < 0.0001$, Table II). Because tabular trees lack spatial awareness of aerodynamic wake advection propagating at 8–12 m s⁻¹ across turbine rows, they cannot represent multi-turbine spatio-temporal phase shifts over 1-hour lead times.

At immediate horizon $h = 1$, Joint Dense Head and Joint Routed exhibit statistical parity across Clean, Noise, and Markov regimes, where bootstrap 95% confidence intervals cross zero (Supplementary Table A11-h2, e.g., +29,624 kW $\cdot$ h, 95% CI [-53,774, +113,023], $p = 0.380$ in Clean). In Delay-6, Joint Routed holds a modest cost advantage (+79,637 kW $\cdot$ h, 95% CI [+7,055, +152,218], uncorrected $p = 0.038$, non-significant under Bonferroni $\alpha = 0.0125$) while matching empirical violation rates (10.6% $\pm$ 1.1% vs. 10.5% $\pm$ 0.5%). Across both horizons, unadapted physical rules exhibit shared vulnerability to telemetry staleness, collapsing under Delay-6 to 24.0% $\pm$ 1.7% violation at $h = 1$ and 12.20% $\pm$ 1.25% at $h = 6$, confirming that aerodynamic power curves require dynamic state adaptation under communication latency.

C. Multidimensional Operational Boundaries and Selective Decision Abstention

To delineate the operational boundaries governing telemetry degradation, we scan across transmission delays $\tau \in \{0, 10, 20, 30, 60\}$ min, horizons $h \in \{1, 3, 6\}$, and pitch channel observabilities (all vs. no_pitch) across all 5 seeds on the 134-turbine WTB plant (Table III). This empirical trajectory delineates a **Three-Regime Operational Framework**:

- 1. Regime I: Simple Recalibration Sufficient ($\tau = 0$ min or mild delays under full observability):** Under fresh telemetry ($\tau = 0$, all), deterministic continuous physical quantiles achieve the lowest operational reserve cost (593,258 $\pm$ 74,886 kW $\cdot$ h at $h = 1$ across the phase-scan sweep, aligning within 0.6% of the primary 589,535 kW $\cdot$ h benchmark, with well-calibrated 7.29% $\pm$ 1.42% violation). Machine-learning representations incur reserve penalties without reliability gains (663,347 kW $\cdot$ h for modular frozen backbones, +11.8%). When telemetry latency increases to $\tau \in [10, 30]$ min with all channels observable, clean-calibrated physical rules collapse (11.18% $\rightarrow$ 13.98% $\rightarrow$ 16.58%). However, state-conditional recalibration fully restores physical compliance (7.1%–7.5% violation) at lower cost than neural models (742,211–939,190 kW $\cdot$ h vs. 786,366–987,760 kW $\cdot$ h), demonstrating that simple recalibration is sufficient when all telemetry channels remain observable.
- 2. Regime II: Learned Representation Advantage (Unobservable pitch channels / partial observability):** When blade-pitch telemetry is withheld across aggregator or OEM bound-

aries (no_pitch), physical aerodynamic rules lose direct rotor state awareness. Recalibrated physics based solely on wind-speed binning inflates reserve costs (824,393 kW $\cdot$ h at $\tau = 0$; 928,028 kW $\cdot$ h at $\tau = 10$; 1,010,530 kW $\cdot$ h at $\tau = 20$; 1,101,292 kW $\cdot$ h at $\tau = 30$; 1,378,900 kW $\cdot$ h at $\tau = 60$). In contrast, learned representations (*Frozen Backbone + Residual Quantile*) reconstruct operating states from electromechanical transients (active/reactive power, rotor speed), achieving 755,794 kW $\cdot$ h at $\tau = 0$ (an 8.3% cost reduction of 68,599 kW $\cdot$ h over recalibrated physics) and maintaining consistent cost savings across all latencies (866,753 kW $\cdot$ h at $\tau = 10$, -61.3k; 947,869 kW $\cdot$ h at $\tau = 20$, -62.7k; 1,054,592 kW $\cdot$ h at $\tau = 30$, -46.7k; 1,330,318 kW $\cdot$ h at $\tau = 60$, -48.6k).

- 3. Regime III: Compound Breakdown and Conservative Safeguards ($\tau = 60$ min, Delay-6):** Under 60-minute latency, unadapted physical rules experience systematic collapse (24.02% $\pm$ 1.69% violation, incurring 127.6 MWh of unhedged shortfall exposure). Condition-matching recalibration restores fleet-average violation below the 10.0% target across all three model families (9.46% for physical rules, 9.55% for Frozen Backbone, and 9.62% for Joint Routed), with the recalibrated physical rule delivering the lowest operating cost (1,228,609 kW $\cdot$ h). This directly refutes the prior assumption that multi-step SCADA latency unconditionally collapses all models or makes selective rejection mandatory to meet fleet-average compliance. However, fleet-average compliance conceals critical tail risk: across the 5 random seeds, every method achieves strictly a 3/5 (60%) seed compliance rate, as Seeds 201 and 203 experience residual breaches (10.2%–10.7%) across all architectures. In this severe telemetry regime, unconstrained neural optimization is vulnerable to tail risk.

TABLE III

MULTIDIMENSIONAL OPERATIONAL PROGRESSION ACROSS TELEMETRY LATENCY $\tau \in [0, 60]$ MIN AND PITCH OBSERVABILITY ($h = 1$, 5-SEED MEAN ON WTB; NOMINAL $\tau = 0$ BASELINE 593,258 kW $\cdot$ h ALIGNS WITHIN 0.6% OF THE PRIMARY FROZEN 589,535 kW $\cdot$ h BENCHMARK). COST IN kW $\cdot$ h ($\rho = 10$).

τ (min)	Pitch	Clean Physical	Recal. Physical	Frozen Backbone	Joint Routed	Operational Regime
		Cost / Viol.	Cost / Viol.	Cost / Viol.	Cost / Viol.	
0	All	593,258 (7.3%)	593,258 (7.3%)	663,347 (6.1%)	663,915 (7.3%)	Regime I (Physics Opt.)
0	None	961,024 (3.2%)	824,393 (5.4%)	755,794 (5.8%)	756,257 (6.4%)	Regime II (Repr. Adv.)
10	All	753,606 (11.2%)	742,211 (7.4%)	786,366 (6.7%)	789,094 (7.6%)	Regime I (Recal. Suff.)
10	None	1,008,743 (4.8%)	928,028 (6.2%)	866,753 (6.1%)	874,156 (6.6%)	Regime II (Repr. Adv.)
20	All	896,198 (14.0%)	841,734 (7.1%)	875,838 (7.4%)	879,750 (8.0%)	Regime I (Recal. Suff.)
20	None	1,062,646 (6.1%)	1,010,530 (6.2%)	947,869 (6.7%)	959,934 (7.1%)	Regime II (Repr. Adv.)
30	All	1,069,266 (16.6%)	939,190 (7.5%)	987,760 (7.1%)	990,091 (7.6%)	Regime I (Recal. Suff.)
30	None	1,138,774 (7.5%)	1,101,292 (6.7%)	1,054,592 (6.4%)	1,064,238 (6.7%)	Regime II (Repr. Adv.)
60	All	1,656,285 (24.0%)	1,228,609 (9.5%)	1,283,855 (9.6%)	1,274,047 (9.6%)	Regime III (Abstain / Inflat.)
60	None	1,437,280 (11.8%)	1,378,900 (9.0%)	1,330,318 (8.7%)	1,331,840 (8.8%)	Regime II (Repr. Adv.)

[†]Exceeds nominal 10% violation target ($q^* = 0.90$). Bold indicates lowest compliant operational cost. Under $\tau = 60$ min, by-seed compliance is 3/5 seeds (60%) across models.

To safeguard operations in Regime III under communication stress, analogous to wide-area power system controllers where telemetry delays trigger watchdog-governed fallbacks [10] or anomaly mitigations [11], we evaluate a **Selective Decision Abstention (Risk-Coverage Policy)**:

$$r_{i,t}^* = \begin{cases} \hat{r}_{i,t}^{\text{learned}}, & \text{if } U_{i,t} \leq \theta_c \quad (\text{Accepted Prediction}), \\ r_{i,t}^{\text{fallback}}, & \text{if } U_{i,t} > \theta_c \quad (\text{Decision Abstention / Refusal}), \end{cases} \quad (15)$$

where $U_{i,t}$ denotes predictive uncertainty (entropy or quantile spread), and θ_c is calibrated on validation data to target coverage fraction $c \in [0.5, 1.0]$. The fallback reserve $r_{i,t}^{\text{fallback}}$ is defined by the conservative envelope:

$$r_{i,t}^{\text{fallback}} = \max \left(r_{i,t}^{\text{phys, recal}}, [P_{\text{rated}} - \hat{P}_{i,t}]_0^{P_{\text{rated}}} \right), \quad (16)$$

where $[x]_0^{P_{\text{rated}}} = \min(\max(x, 0), P_{\text{rated}})$ clamps available generation headroom against rated nameplate capacity ($P_{\text{rated}} = 1,500$ kW), lower-bounded by the condition-recalibrated aerodynamic quantile rule $r_{i,t}^{\text{phys, recal}}$. To audit this mechanism, Table IV benchmarks selective abstention against two mandatory controls: **Random Margin Inflation** (evaluated both as an ex-post equal-budget diagnostic benchmark matching total fleet reserve MWh, and as an ex-ante deployable policy with inflation multiplier α_{val} calibrated

on validation data).

TABLE IV

RISK-COVERAGE SELECTIVE DECISION ABSTENTION VS. RANDOM AND UNIFORM CONTROLS UNDER DELAY-6 ($\tau = 60$ min, $h = 1, 5$ -SEED AGGREGATE ON WTB).

Model	Target c (%)	Selective Abstention Fleet Viol. / Cost	Random Control Fleet Viol. / Cost	Uniform Inflation Fleet Viol. / Cost
<i>Mode: All Channels Observable</i>				
Joint Routed	1.00 (1.00)	9.62% / 1,274,047	9.62% / 1,274,047	9.62% / 1,274,047
	0.90 (0.94)	9.44% / 1,274,784	9.50% / 1,272,870	9.45% / 1,273,944
	0.80 (0.85)	9.14% / 1,277,532	9.36% / 1,270,990	9.24% / 1,274,781
	0.50 (0.53)	8.54% / 1,276,252	8.78% / 1,265,902	8.80% / 1,277,570
<i>Mode: No Pitch Telemetry (Withheld)</i>				
Frozen Backbone	1.00 (1.00)	9.52% / 1,283,268	9.52% / 1,283,268	9.52% / 1,283,268
	0.90 (0.94)	9.30% / 1,277,806	9.36% / 1,281,737	9.36% / 1,283,424
	0.80 (0.85)	9.01% / 1,273,271	9.28% / 1,279,718	9.16% / 1,284,225
	0.50 (0.53)	8.37% / 1,264,384	8.76% / 1,271,385	8.78% / 1,286,221
<i>Mode: No Pitch Telemetry (Withheld)</i>				
Frozen Backbone	1.00 (1.00)	8.68% / 1,331,356	8.68% / 1,331,356	8.68% / 1,331,356
	0.90 (0.92)	8.93% / 1,343,555	8.61% / 1,334,006	8.83% / 1,330,829
	0.80 (0.84)	9.15% / 1,350,059	8.66% / 1,337,746	8.92% / 1,331,452
	0.50 (0.55)	8.84% / 1,371,729	8.55% / 1,353,405	8.29% / 1,333,906

Cost in kW·h ($p = 10$). Fleet target $q^* = 0.90$ (10% viol). Note: In *No Pitch*, uniform margin inflation strictly Pareto-dominates selective abstention: ex-post equal-budget uniform inflation achieves 8.29% violation at 1,348 kW·h, and ex-ante validation-frozen inflation achieves 7.12% violation at 1,348 kW·h (both lower than selective abstention at 8.84% and 1,372 kW·h).

The empirical audit reveals that **heuristic uncertainty scores (predictive entropy and quantile spread) fail to reliably isolate tail shortfall risk under severe telemetry staleness**. Under observable telemetry (all), target coverage $c = 0.90$ yields a realized empirical coverage of $\hat{c} = 93.42\% \pm 0.81\%$, but selective abstention reduces fleet violation only marginally (9.62% $\rightarrow$ 9.44% for Joint Routed; 9.52% $\rightarrow$ 9.30% for Frozen Backbone). This reduction is statistically indistinguishable from random abstention (9.50% and 9.36%, $p > 0.40$), while by-seed compliance remains strictly 3/5 seeds (60%), as Seed 201 accepted violation actually rises to 10.48% (vs. 10.22% unrejected). Under withheld pitch (no_pitch), selective abstention is strictly Pareto-dominated by uniform margin inflation: at $c = 0.50$, selective abstention costs 1,371,729 kW·h (8.84% viol), whereas ex-post test-matched uniform margin inflation achieves lower violation (8.29%) at lower total cost (1,333,906 kW·h, saving 37,823 kW·h), and ex-ante validation-frozen inflation achieves 7.12% violation at 1,348,361 kW·h (saving 23,368 kW·h). Because stale SCADA telemetry corrupts both the point prediction and the uncertainty proxy, selective rejection incurs the penalty of bloated fallback reserves on safe turbines while missing true tail shortfalls. In industrial operations facing multi-step SCADA disruption, **uniform reserve margin inflation and aerodynamic physical fallbacks provide more reliable, cost-effective risk hedging than heuristic selective abstention**.

D. Operating-Boundary Recovery and Point-Forecast Price of Routing

Evaluating point-forecasting accuracy across five WTB seeds, the boundary-forced router achieves an overall RMSE of 236.13 $\pm$ 8.41, compared with 224.34 $\pm$ 2.23 for iTransformer and 225.74 $\pm$ 2.60 for Graph WaveNet (a margin of 11.79 units vs. iTransformer and 10.39 units vs. Graph WaveNet). Provenance-corrected train-only class weighting narrows this gap to 5.59 units (reaching 229.93 $\pm$ 2.50). While unconstrained models optimize solely for whole-sample RMSE, they lack operating-state awareness and suffer severe reliability degradation under telemetry impairments. In contrast, the boundary-forced router preserves physical state alignment, achieving NMI 0.8716 $\pm$ 0.0418 and ARI 0.9166 $\pm$ 0.0371 under strict masking (NMI 0.721 and ARI 0.740 under train-only supervision).

Input ablation stress guards confirm that routing fidelity does not arise from target leakage: withholding Patv or Pab_mean preserves mean NMI at 0.881 and 0.878, while lagging Patv and lagging pitch/wind yield 0.763 and 0.684. In addition, gate matching to the active operating regime peaks at the transition step (0.815) and attenuates under three- and six-step lead shifts (0.361 and 0.405), confirming that gate activations track dynamic physical state transitions.

E. Withheld-Channel Electromechanical Signature Recovery

The operational integrity of the blind-spot fallback rests on whether the model can infer operational regime boundaries when primary pitch and anemometer channels are withheld across commercial OEM protocol boundaries or VPP administrative interfaces. To test whether secondary electromechanical consequence channels

carry sufficient information to reconstruct operating state posteriors without circular dependence, we formulate a strict **counterfactual stress-testing probe**: withholding Wspd and Pab_mean while retaining active power and downstream electromechanical channels yields a mean NMI of 0.561 and ARI of 0.635 across five seeds on WTB (Table V). Removing active power (signature_core) drops mean NMI to 0.367, whereas permuted label negative controls collapse to 4.0×10^{-6} , definitively ruling out spurious correlations.

TABLE V

SIGNATURE-GATE IDENTIFIABILITY PROBE ACROSS COMMERCIAL WIND FARMS UNDER HETEROGENEOUS PITCH OBSERVABILITY. DEFINING CHANNELS ARE WITHHELD FROM THE GATE ANCHOR; ALL VALUES ARE MEAN $\pm$ SD OVER 5 SEEDS.

Probe	RMSE	NMI / ARI	Interpretation
Canonical full-anchor signature_full (no Wspd/Pab)	229.93 241.84 $\pm$ 7.19	0.7208 / 0.7398 0.5613 $\pm$ 0.0199	Train-only clean baseline Boundary recoverable from conseq. channels
signature_core (no Wspd/Pab/Patv)	302.10 $\pm$ 9.16	0.3671 $\pm$ 0.0372 0.0239 0.4342 $\pm$ 0.0378	Weaker non-power signature; 3/5 seeds collapsed
sig_full_shuffled	241.50 $\pm$ 10.25	4.0e-6 $\pm$ 2.1e-6 -1.3e-5 $\pm$ 5.3e-5	Negative control: chance level under permuted labels
Unconstrained MoE	—	0.0147	No declared-boundary supervision
<i>Cross-farm replication, ENGIE La Haute Borne (4 turbines, 99% pitch)</i>			
LHB canonical (full anchor)	185.0 $\pm$ 1.9	0.9752 $\pm$ 0.0088 0.9904 $\pm$ 0.0044	Full-anchor baseline, identical protocol
LHB signature_full	185.1 $\pm$ 1.4	0.6743 $\pm$ 0.1209 0.7401 $\pm$ 0.1604	Boundary signature transfers; min-seed 0.499
LHB signature_core	188.0 $\pm$ 1.0	0.5750 $\pm$ 0.0423 0.6270 $\pm$ 0.0638	Non-power signature transfers; 0/5 collapsed
<i>Commercial UK wind plants (Penmanshiel MM82, Kelmarsh MM92)</i>			
Penmanshiel signature_full	—	0.3024 $\pm$ 0.0343	Above the 0.20 criterion; shuffled 4e-5
Penmanshiel signature_core	—	0.1954 $\pm$ 0.0961	Borderline: below 0.20 but 4400x above shuffled chance
Kelmarsh signature_full	—	0.3402 $\pm$ 0.0736	Above the 0.20 criterion; shuffled 9.2e-5
Kelmarsh signature_core	—	0.3775 $\pm$ 0.0783	Above the 0.20 criterion

To prevent over-generalization and evaluate external deployment boundaries, we strictly disentangle three disparate lines of external empirical evidence across commercial plants:

- Within-plant local retraining (ENGIE La Haute Borne & UK dual-farm replication):** Models trained and evaluated locally under site-specific rated aerodynamic parameters demonstrate that the neural architecture adapts to distinct turbine technologies, confirming that consequence signatures are recoverable from electromechanical transients without direct pitch observation:
 - ENGIE La Haute Borne (4 turbines, 99% pitch, $v_{\text{rated}} = 14.5 \text{ m s}^{-1}$):** Withheld-channel probe retains an NMI of 0.674 for signature_full and 0.575 for signature_core (collapsing to 1.92×10^{-4} under permuted controls; five-seed routing NMI 0.941, ARI 0.971).
 - Kelmarsh (6 Senvion MM92 turbines, $v_{\text{rated}} = 12.5 \text{ m s}^{-1}$):** With defining wind-speed and pitch-angle channels withheld, signature_full achieves an NMI of 0.3402 ± 0.0736 (3700x above shuffled chance). In 5-seed local chronological dispatch replay at $h = 6$, locally retrained Joint Routed yields $70,375 \pm 16,324 \text{ kW} \cdot \text{h}$ (Clean) and $109,359 \pm 13,634 \text{ kW} \cdot \text{h}$ (Delay-6), outperforming GBDT ($+44,891 \text{ kW} \cdot \text{h}$ and $+29,621 \text{ kW} \cdot \text{h}$, $p < 0.05$) and continuous physical quantiles ($82,456 \text{ kW} \cdot \text{h}$ and $116,409 \text{ kW} \cdot \text{h}$).
 - Penmanshiel (15 Senvion MM82 turbines, $v_{\text{rated}} = 14.5 \text{ m s}^{-1}$):** With defining channels withheld, signature_full achieves 0.3024 ± 0.0343 (6800x above shuffled chance). In local dispatch replay at $h = 6$, locally retrained Joint Routed yields $831,251 \pm 112,331 \text{ kW} \cdot \text{h}$ (Clean) and $813,831 \pm 102,946 \text{ kW} \cdot \text{h}$ (Delay-6), versus GBDT ($1,064,962 \text{ kW} \cdot \text{h}$ and $1,086,102 \text{ kW} \cdot \text{h}$, $p < 0.001$), where GBDT experiences severe violation inflation (24.7% to 26.5%). These empirical findings demonstrate model architecture adaptability to site-specific aerodynamic rated parameters under local retraining, rather than uncalibrated zero-shot cross-farm generalizability.

- Directional transfer boundaries across disparate wind plants:** In contrast to local chronological retraining, di-

rect zero-shot cross-farm transfer without site-specific recalibration reveals pronounced directional transfer sensitivity. As audited by our pre-registered external wind guard (`external_wind_guard.json`), transferring representations from Kelmarsh to Penmanshiel preserves moderate boundary alignment (NMI 0.752–0.770), whereas reverse transfer from Penmanshiel to Kelmarsh yields NMI 0.341–0.505 (pooled bidirectional mean NMI 0.557). This directional asymmetry demonstrates that aerodynamic operating boundaries are intimately coupled to turbine rotor dimensions and local wake topographies, establishing that reliable multi-farm deployment requires site-specific sensor calibration and local retraining.

3. **Longitudinal multi-year operational stability:** Across decadal longitudinal archives (Kelmarsh, 9 years; Penmanshiel, 8.6 years), walk-forward evaluations examine periodic quantile recalibration under multi-year climatological drift and turbine component aging. These longitudinal evaluations demonstrate the operational mechanics of periodic schedule recalibration across multi-year operating horizons, establishing empirical baselines for long-term fleet management under environmental drift.

F. Degradation Resilience and Early Warning Dynamics

Table VI evaluates detection fidelity under fair input degradation (Unified Arrival Layer), wherein full-stream history shifts and anchor delays are applied symmetrically to all models. Under clean telemetry, deterministic physical thresholds exhibit superior sensitivity (recall 1.000 vs. 0.9705), while logistic regression achieves 1.000 recall with 0.879 precision (vs. 0.630 for the routed posterior). Under 6-step latency ($d = 6$, 60-min history stall), however, deterministic rule recall collapses from 1.000 to 0.196 (0.249 F1). In contrast, the spatio-temporal boundary posterior sustains an early-window recall of 0.4170 ± 0.1137 (+0.221 gain, more than doubling detection sensitivity) with an F1 score of 0.3802 (+0.131 gain). While 60-minute communication latency naturally attenuates absolute detection recall ($0.971 \rightarrow 0.417$), the dynamic wake graph doubles surviving operational resilience without look-ahead leakage.

TABLE VI
EARLY MPPT-TO-PITCH DETECTION UNDER FAIR INPUT DEGRADATION (UNIFIED ARRIVAL LAYER) ACROSS 5 TRAIN-ONLY SEEDS. FULL-STREAM HISTORY SHIFT AND ANCHOR DELAY APPLIED SYMMETRICALLY TO GATE AND RULE.

Scenario	Setting	Gate	Rule	Gain
Clean anchors	—	0.9705 $\pm$ 0.0299	1.000	−0.029
Delay (fair)	1 step	0.8346 $\pm$ 0.0771	0.655	+0.179
Delay (fair)	3 steps	0.6200 $\pm$ 0.1209	0.381	+0.239
Delay (fair)	6 steps	0.4170 $\pm$ 0.1137	0.196	+0.221
Noise (fair)	Wspd 0.5, Pab 1.0	0.9635 $\pm$ 0.0375	0.811 $\pm$ 0.016	+0.152
Noise (fair)	Wspd 1.0, Pab 2.0	0.9505 $\pm$ 0.0363	0.680 $\pm$ 0.019	+0.271
Availability	75% labels	0.971	0.744	+0.227
Availability	50% labels	0.971	0.508	+0.462
Availability	25% labels	0.971	0.242	+0.729

Under industrial two-state Markov-Gilbert burst dropouts ($p_{GB} = 0.08$, $p_{BB} = 0.75$, $d \leq 6$) evaluated symmetrically across both anchor and encoder history channels, the stale threshold rule’s recall drops to 0.840 with an F1 of 0.904, whereas the jointly-learned routed posterior sustains a recall of 0.962 ± 0.023 and an F1 of 0.925 ± 0.042 during bursts (mean F1 gain +0.021, 95% bootstrap CI [−0.014, +0.050]). During dynamic transitions, the spatio-temporal graph provides vital defense-in-depth against communication dropouts.

G. Boundary-Window Reserve Vignette and Benchmark Comparison

At a shortage-to-reserve cost ratio $\rho = 10$ on the transition boundary band, the boundary-router gate-bin policy costs 95.13M, outperforming the same-model global rule (99.55M), while the low-RMSE Graph WaveNet with physical-bin reference achieves 84.31M. This translates into approximately 637 avoided MWh-equivalent shortage cells and a 442k illustrative reserve-cost-scale marker at 100 EUR/MWh.

H. Point of Common Coupling (PCC) Portfolio Smoothing & Exploratory Longitudinal Drift Analysis

Aggregating decentralized turbine power flows at the Point of Common Coupling (PCC) bus ($P_{\text{farm}} = \sum_{i \in \mathcal{V}_t} P_{i,t}$, averaging ~ 121 active turbines) unlocks spatial portfolio diversification: localized high-frequency aerodynamic turbulence cancels out across the array, leaving boundary transitions as the dominant source of reserve shortfall risk. Across the full operational envelope, joint posterior aggregate quantile pricing reduces reserve cost by −2.32M kWh (95% bootstrap CI [−6.07M, 1.73M]) against global PCC quantiles and by −1.37M kWh (CI [−6.33M, 2.82M]) against Gaussian parametric sizing; in transitional regimes (10%–90% pitching), it saves −1.33M kWh (CI [−1.68M, −1.07M], strictly excluding zero) over continuous physical pitch, confirming net economic value after fleet-wide cancellation. Interfacing scheduled reserves with a co-located battery energy storage system (BESS, 20,MWh / 10,MW) under receding-horizon Model Predictive Control (MPC) reduces Delay-6 dispatch cost by 32.1% (\$487,746 to \$331,257) and cuts shortage energy by 45.7% (2,514.9 to 1,401.2,MWh; Supplementary Table A11-BESS). These upstream values represent the PSREI risk proxy ($\rho = 10$) rather than wholesale market cashflow settlements.

Across two European commercial wind plants with longitudinal records (Kelmarsh, 9 years, 2016–2024, 6 MM92 turbines; Penmanshiel, 8.6 years, 2016–2024, 15 MM82 turbines), exploratory evaluations examine periodic quantile recalibration under multi-year climatological drift and turbine aging. These longitudinal evaluations illustrate the operational mechanics of periodic schedule recalibration across decadal operating horizons, establishing empirical baselines for long-term fleet management.

V. DISCUSSION

A. Accuracy, accountability and deployment gates

The operational role of the boundary-aware architecture is a specialized diagnostic for reserve risk screening rather than an unconstrained whole-sample point predictor. When bulk energy scheduling operates under pristine SCADA telemetry, unconstrained spatio-temporal architectures (Graph WaveNet, iTransformer) deliver superior aggregate accuracy; conversely, when telemetry is delayed, corrupted, or pitch-withheld, boundary-risk posteriors provide indispensable diagnostic defense. To ensure rigorous external validation, we strictly disentangle three lines of empirical evidence: (a) **Cross-farm directional transfer boundaries:** Direct zero-shot transfer across disparate commercial plants exhibits pronounced directional asymmetry (Kelmarsh $\rightarrow$ Penmanshiel NMI 0.752–0.770, ARI 0.735–0.761 vs. Penmanshiel $\rightarrow$ Kelmarsh NMI 0.341–0.505, pooled bidirectional mean NMI 0.557; `external_wind_guard.json`), confirming that unadapted cross-farm deployment is circumscribed by local aerodynamic geometry, and that reliable multi-site transfer requires site-specific sensor recalibration. (b) **Within-plant local retraining:** Local chronological training across ENGIE La Haute Borne (five-seed routing NMI 0.941, ARI 0.971; canonical NMI 0.975; withheld probe 0.674/0.575), Kelmarsh (probe 0.340/0.378), and Penmanshiel (probe 0.302, `signature_core` 0.195; Supplementary Table A9d) confirms neural architectural adaptability to site-specific aerodynamic rated speeds. (c) **Longitudinal rolling walk-forward evaluation:** Across longitudinal multi-year records (Kelmarsh, 9 years; Penmanshiel, 8.6 years), periodic 2-year sliding window evaluations characterize operational recalibration mechanics under climatological drift, tracking parameter evolution over decadal operating horizons.

External walk-forward rolling evaluation establishes empirical physical boundary conditions for the pre-registered admission protocol (Supplementary Table A14). On ENGIE La Haute Borne (4 turbines, 99% complete pitch), walk-forward pooled quarterly evaluation (Q1–Q3) incurs a positive cost delta of +1.01M kWh (95% bootstrap CI [+0.29M, +1.76M], strictly excluding zero) versus the global quantile and +0.92M kWh (CI [+0.26M, +1.61M]) versus continuous physical pitch (Supplementary Table A11h). This reveals a quantile variance amplification under acute seasonal drift: on a

miniature 4-turbine site without Point of Common Coupling (PCC) spatial portfolio smoothing, short quarterly slices suffer variance spikes during autumn regime shifts (+968.1k kWh in Q3, while Q1–Q2 cross zero). Expanding calibration to a 180-day annual window compresses the cost gap to +43k kWh (CI [−28.7k, +141.2k], strictly crossing zero), bounding soft-posterior reserves to PCC-smoothed plants and degraded/pitch-sparse telemetry.

Evaluating across shortage penalty ratios $\rho \in \{5, 10, 20\}$ articulates an operational envelope (“Telemetry Availability $\times$ Penalty Ladder”). At $\rho = 10$, learned posteriors dominate unconditioned baselines. At $\rho = 20$, asymmetry emerges: at Kelmars, physical rules recover and surpass soft-gate pricing by +0.95M kWh (CI [+0.15M, +1.65M]); at Penmanshiel, the soft gate leads physical rules by −7.51M kWh (CI [−13.51M, −1.25M]) while crossing zero versus the unconditioned global baseline (−5.15M kWh, CI [−12.47M, +2.17M]). Direct reserve deployment at new farms requires pitch or proxy observability, boundary support, compatible geometry, local recalibration, and a held-out validation pass [8].

VII. LIMITATIONS

WTB pseudo-labels derive from wind speed and pitch angle, channels present in full gate anchors. Counterfactual withheld-channel stress testing confirms that the boundary survives without direct pitch observations via consequence signatures across four farms (permuted controls at chance). Removing pitch dispersion retains non-power NMI 0.309 on WTB and 0.578 on La Haute Borne, confirming dispersion does not carry the signal. Yet boundary identification remains an anchor-constrained diagnostic rather than anchor-free discovery [18], [23]. Inputs may contain history $\text{Patv}_{t-H+1:t}$ and issue-time Patv_t , while targets begin at $t + 1$. Expert semantics reflect the declared mapping; unassigned logits are not universal physical states.

The analytical scope of this study is formally bounded to Level 1 local pre-dispatch risk screening proxies (PSREI at $\rho = 10$) designed to mitigate unhedged energy imbalance entering real-time grid balancing. The formulation does not model Level 2 transmission network clearing with Locational Marginal Prices (LMPs), full AC-OPF power-flow constraints, or multi-stage settlement cashflows [29], [31]. Core empirical findings remain five-seed WTB recovery, mechanism ablation, La Haute Borne local replication, and reserve risk triage.

VIII. CONCLUSION

This study exposes the operational vulnerability and reliability breakdown of deterministic wind turbine operating reserve rules under SCADA telemetry degradation and evaluates jointly-learned boundary-risk posteriors as relative mitigators and diagnostic instruments. Across a 5-seed benchmark on the 134-turbine WTB plant, continuous aerodynamic quantile rules achieve the lowest PSREI reserve-screening cost under pristine telemetry (589,535 kW · h at $h = 1$, 881,367 kW · h at $h = 6$), but undergo systematic reliability collapse under a 6-step (60 min) delay, surging violation rates to $24.0\% \pm 1.7\%$ ($h = 1$) and $12.20\% \pm 1.25\%$ ($h = 6$). Under Delay-6, Continuous Physical Quantile reaches $1,136,466 \pm 84,101$ kW · h at $h = 6$ and $1,228,609 \pm 71,940$ kW · h at $h = 1$ under condition-matching recalibration.

Under frozen validation calibration, state-conditional recalibration of physical rules recovers violation to 10.7% and cuts shortage from 127.6 MWh to 57.0 MWh (55.3% reduction), while learned boundary-risk posteriors provide an additional 5.4% reduction to 53.9 MWh (a combined 57.7% reduction) and double regime recall (0.416 vs. 0.196 under Delay-6). However, data-driven representations alone do not provide unconditional fail-safe operation: under 6-step latency, empirical violation rates settle at $10.6\% \pm 1.1\%$, demonstrating that learned models substantially alleviate, but cannot completely eliminate, reserve breakdown without conservative margin safeguards. Capacity-matched Dense and MoE heads achieve statistical parity in Clean (cost ratio 1.045, $p = 0.380$) with a modest Delay-6 margin (ratio 1.056, uncorrected $p = 0.038$), while modular architectures match tail reliability (9.7% violation), indicating that operational value originates from shared spatio-

temporal representations rather than dynamic routing. Direct zero-shot cross-farm transfer exhibits directional asymmetry (Kelmars $\rightarrow$ Penmanshiel NMI 0.752–0.770 vs. Penmanshiel $\rightarrow$ Kelmars 0.341–0.505, pooled 0.557), bounding reliable deployment to sites with local retraining (ENGIE La Haute Borne, NMI = 0.941). These findings reframe data-driven reserve models into rigorous, accountable diagnostic tools that delineate the physical and communication boundaries of smart grid operations.

IX. AI USE STATEMENT

The authors used OpenAI ChatGPT/Codex only for language editing, consistency checks, and submission-material drafting; all data, analyses, references, conclusions, and final text were reviewed and controlled by the authors.

X. CODE AND DATA AVAILABILITY

The raw KDD Cup 2022, ENGIE La Haute Borne, Kelmars, and Penmanshiel SCADA datasets are public (raw third-party data not redistributed). Code, configurations, releasable derived tables, figure data, and checkpoints will be made available with the article. The reproduction package covers WTB routing, reserve audit, anchor-stress caches, early-warning label degradation, classifier control, class-weight sensitivity, external diagnostics, and evidence-freeze protocols across declared seeds. Results are statistically reproducible across declared seeds rather than bitwise deterministic across all GPU/CUDA environments. The analysis involves no human subjects.

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